C Programming

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How To Learn Programming

- Learn the Concept in Lecture
- Understand the program flow and concept in lecture
- **“Practice the code in lab by typing the code in lab by your own”**
- Discuss with your friend and teach them.

Agenda

- C Programming Subject Contents
- C development environment
- Introduction to C
- Control flow
- Functions
- Storage class
- Pointer
- Type qualifiers
- Preprocessor Directives
- Array
- Dynamic memory allocation
- Structure & union
- Enum
- File IO
- Function pointer
- Bitwise operator and Advanced features

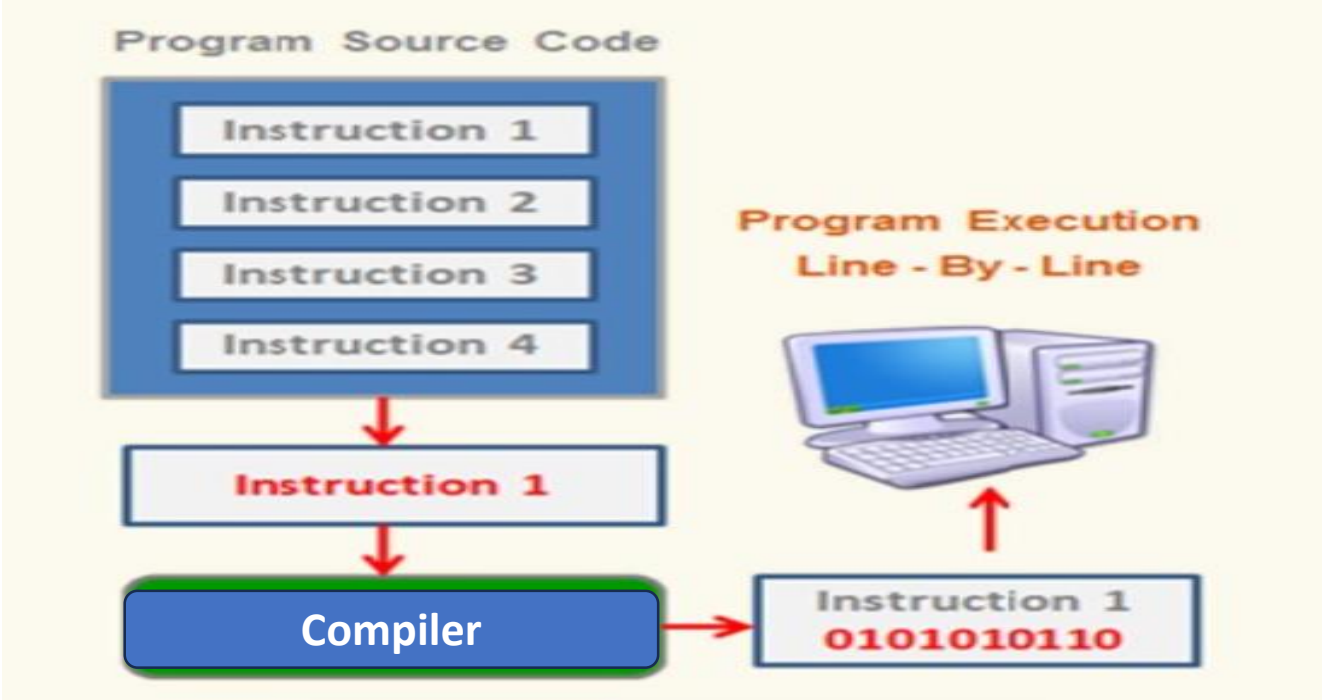
Computer and Program

What is Computer?

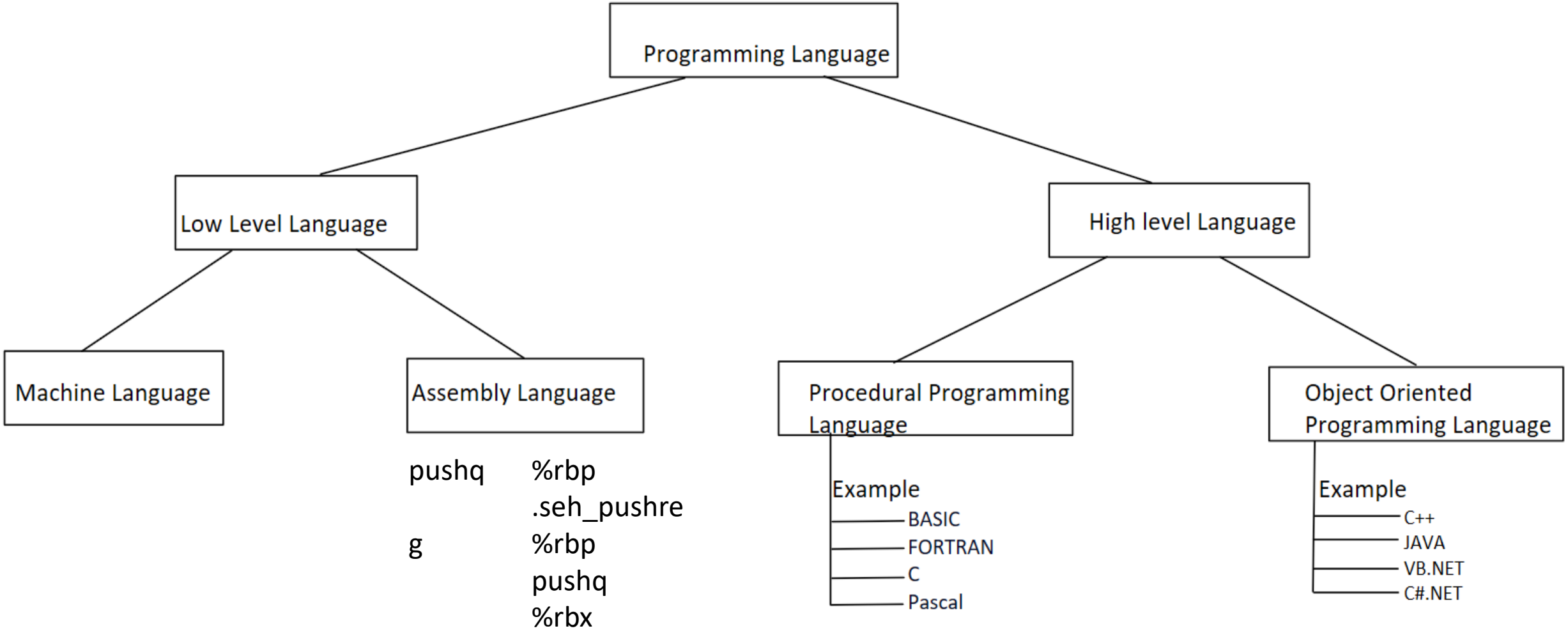
- It is a machine/hardware/digital device which does various tasks for the user efficiently and effectively.

What is Program?

- Set of instructions given to the machine to do specific task.



Classification of Languages



Classification of Languages

- **Low Level Language**
  - The low level language is a programming language where machine instructions can be given **in 0 and 1 form**.

- **Machine Level Language**
  - The machine-level language is a language that consists of a **set of instructions that are in the binary form 0 or 1**. As we know that computers can understand only machine instructions, which are in binary digits, i.e., 0 and 1, so the instructions given to the computer can be only in binary codes.
    - **Performance is good** as we are directly writing the program on machine
    - **Platform dependent**
    - **Difficult to program**
    - **Error prone**

Classification of Languages

- **Assembly Language**
  - The assembly language contains **some human-readable instructions** (**consists of numbers, symbols, abbreviations**), The language was introduced in **1952.**
  - As we know that **computers can only understand the machine-level instructions**, so we require a translator that converts the assembly code into machine code. The **translator used** for translating the code is known as **an assembler.**
    - Easier to understand and use and to locate errors
    - **Platform dependent**
    - Knowledge of hardware will help to program Machine level coding
    - Execution is slower compared to machine level language.
- **High Level Language**
  - Make **easy communication** to computer system
  - **Independent to particular type of computer**
  - More **close to human language** than machine language.
    Compiler helps to convert instructions to machine understandable form.
  - Takes **additional time to translate the instructions to machine** understandable instructions.

History Of C Programming

- C language was developed **by Dennis Ritchie in 1972 at AT & T Bell Labs** on PDP-11 (Programmed Data Processor) machine.

- It was developed while porting UNIX from PDP-7 to PDP-11.

- Many features of C are inspired from B language (Ken Thompson) and BCPL (Martin Richards).

- Initial release of C is referred as K & R C.

Standardization

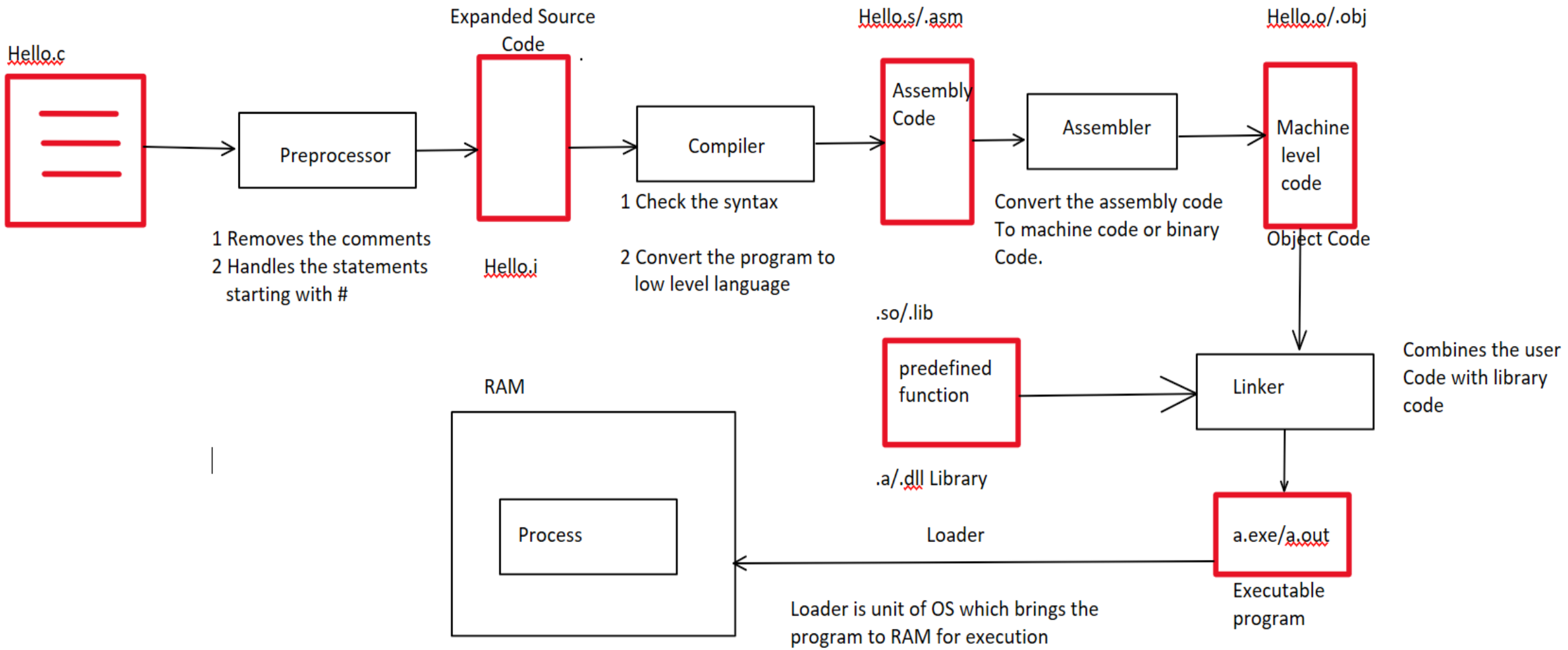
- C was standardized by **ANSI in 1989**. This is referred as C89.
- Standardization ensures C code to **remain portable**.
- C standard is revised **multiple times** to **add new features** in the language.
- C89 – First ANSI standard
- C90 – ANSI standard adopted by ISO
- C99 – Added few C++ features like bool, inline, etc.
- C11 – Added multi-threading feature.
- C17 – Few technical corrections.

Compilation & Execution of C program

- **Preprocessor:-**
  - A preprocessor accepts the inputs in source language and produces output source program that is acceptable to the compiler.
  - Main task of preprocessor is to remove the comments and handle all the statements starting with #

- **Linker:-**
  - Single programmer can write small program and store it in single source code file , However **Large size software consist of several thousands and several millions of code** , to take care of this approach software developers generally follow the **modular approach**.
  - In modular approach software consist of multiple source file , In this case we use the software called as linker to combine all object programs and to convert into final executable.
  - **Linker is a software** that takes multiple object files and fits them together to assemble into **final executable**.

Compilation & Execution of C program



Toolchain & IDE

- Toolchain is set of tools to convert high level language program to machine level code.
  - Preprocessor
  - Compiler
  - Assembler
  - Linker
  - Debugger
  - source code editor
  - build tools, etc.
- Popular compiler (toolchains)
  - GCC
  - Visual Studio
- IDE – Integrated development environment
  - Visual Studio
  - Eclipse
  - VS Code (+gcc), etc

Strengths of C Language

- Low-level nature **makes it very efficient** (pointers, data structures)
- Language itself is **very readable** (macros, enum, functions, …)
- **Many compilers available** for a wide range of machines and platforms
- Standardization improves **portability of code**
- Efficient in terms of coding effort needed for a wide variety of programming applications**. Can access OS features** (functions/commands)
- **Effective memory access** (bitwise operators, bit-fields, unions)
- **Extensive library functions** (math, strings, file IO, …)

Applications

- System programming
- OS development
- Device drivers
- System utilities
- Embedded programming
- IoT development
- Language development
- Compiler development
- Achievements (tiobe.com)
- In top-2 languages in last 40 years.
- Language of year: 2019, 2017, 2008.

Hello World

Source Code

```
// Hello World program
#include <stdio.h>
int main() {
printf("Hello World\n");
return 0;
}
```

- Commands

To compile the code

- cmd> gcc hello.c

To execute the code

- cmd> .a.exe

Hello World

- printf() – library function
- stdio.h – header file
- main() – entry point function
- void main() { … }
- int main() { … }
- int main(void) { … }
- int main(int argc, char *argv[]) { … }
- int main(int argc, char *envp[]) { … }
- return 0 – exit status

Algorithm

- **Algorithm - Logical flow** mentioned for given problem statement

Algorithm can be represented in two ways:

- **1. Pseudocode -** Logical flow represented using simple english language
- **2. Flowchart - pictorial representation of logical flow.**
- Accept two numbers from user and display addition of them.

Pseudocode and Flowchart

- **Pseudocode**
- Start

- Declare num1,num2,sum

- Display "Specify number"
- Accept num1

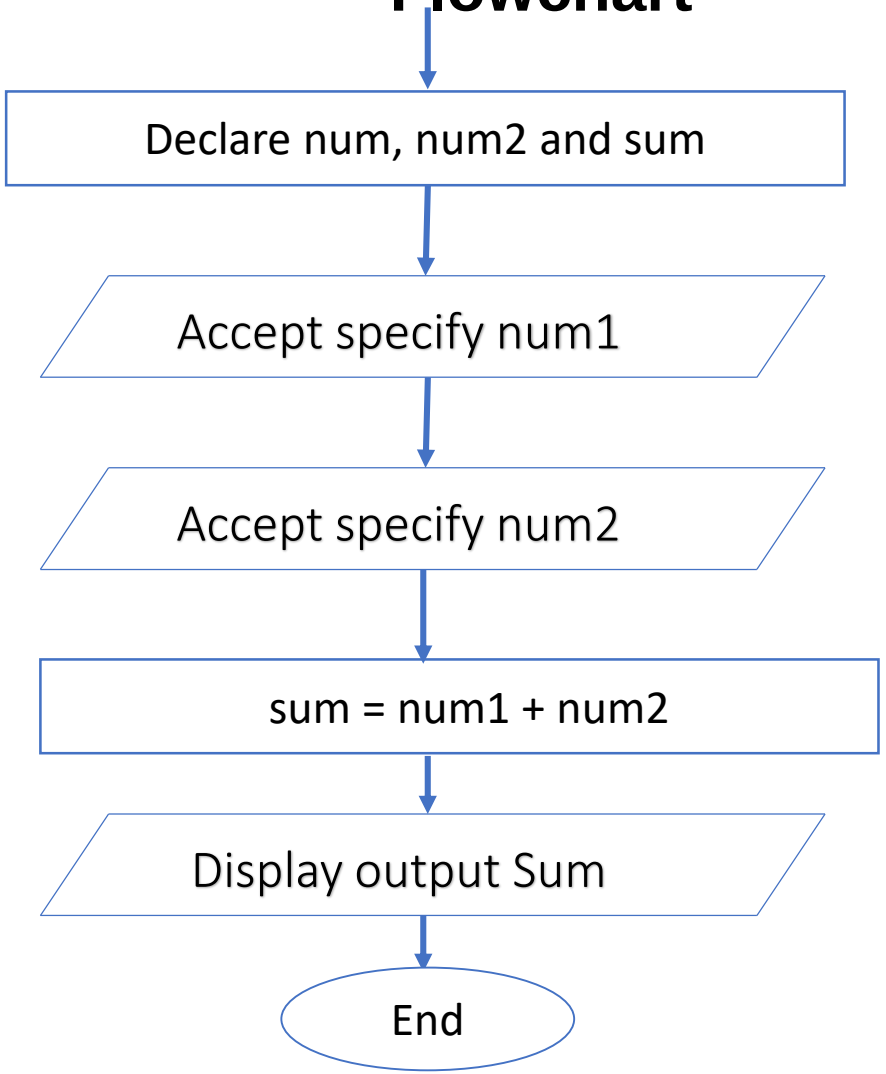
- Display "Specify number"
- Accept num2

- sum = num1 + num2

- Display "Sum is",sum

- end

Flowchart



Thank You